SSI: Building Compliant Power Elements for Servers

Brent R. Mower

SSI Program Manager
Intel Corporation
Enterprise Server Group
15400 NW Greenbriar Parkway
Beaverton, Oregon 97006

***Abstract*-The goal of the SSI (Server System Infrastructure) initiative is to deliver a set of open industry specifications covering two primary server elements: power supplies and electronic bays, and thus lay the groundwork for future growth in the server market. The initiative is similar to such desktop PC efforts as ATX and NLX in that it is working to define common packaging elements for the server market.**

I. INTRODUCTION

During the 1990's, servers have undergone a fundamental change in price/performance and functionality, just as PC's did in the 1980's. The rapid pace of server technology redefined the Intel architecture server from desktop PCs deployed as servers to robust, purpose built, multiprocessing systems. Increases in microprocessor performance and functionality drive corresponding increases in memory, I/O, storage and power and thermal requirements for servers based on the Intel Architecture.

To accommodate these changes, manufacturers developed a new server chassis for each generation of technology. With each new chassis came a set of unique system elements that could share a common form factor. Instead, each new server is designed from ground up since no server standards exist for system designers and industry suppliers. This process forces longer development cycles, greater R&D cost, and low re-use of prior design.

The lack of a standard specification for common server system elements requires server system vendors to frequently redesign the basic infrastructure, such common server system elements as the power supply and chassis, of their server products—a proprietary, expensive and time-consuming process. Intel is working with leading server OEMs and industry suppliers to help remedy the situation.

Unfortunately, proprietary designs for these common elements add little value to end-user solutions. Due to lower volumes and manufacturing inefficiencies, frequently redesigned server elements generally fail to reduce overall costs. In addition, proprietary solutions limit the available supplier base and hamper supplier-driven product innovation. For server system vendors and their suppliers, the server element redesign process is expensive and time consuming, leading to longer product development and time-to-market cycles.

Taking Initiative

In light of these issues, Intel is cooperating with leading server vendors and industry suppliers to lead the way with an initiative to define the common elements of today's and tomorrow's server system infrastructures.

Called the SSI (Server System Infrastructure) initiative, the effort is designed to help the industry develop a common set of server element specifications, and thus lay the groundwork for future growth in the server market. The SSI initiative will:

- Extend the longevity of server packaging chassis into future generations
- Reduce OEM and IHV R&D expenses related to system integration
- Provide a foundation for interchangeable server subsystems
- Broaden the supply base for server system elements, leading to reduced costs
- Continue to advance the award-winning, price/performance leadership position of Intel Architecture-based servers

The SSI initiative's goal is to deliver a set of specifications covering two primary server elements: power supplies and electronics bays. The latter term refers to the physical space in the server chassis allotted to the motherboard—including the microprocessor, memory, I/O

0-7803-5160-6/99/$10.00 © 1999 IEEE.

and related chip sets for entry-level servers (in high-end servers, I/O functions are typically handled off the system motherboard).

For power supplies, the SSI specification addresses such factors as physical dimensions, wattage range and electromechanical interface parameters. In the case of electronics bays, the specification will cover such parameters as size and space, cooling functions, and even physical mounting considerations. The SSI initiative calls for developing a set of specifications that address all segments of the server market, from entry-level products on up through advanced enterprise servers.

In its final form, the SSI specification will define two stacks of common server elements and a set of system design guides (see Table 1). One stack will contain a series of power supplies and the other a series of electronics bays addressing each segment of the IA server market.

User Benefits

While the SSI Initiative is not being promoted as an end-user oriented program, several benefits of the initiative may none-the-less accrue to end-users over time. Common specifications for computing devices, systems and interfaces provide multiple benefits to end-users:

- Reduced time span from technology advancement to commercial availability.
- Faster enterprise configuration and integration projects.
- Lower product cost as multiple manufacturers compete in an open market based on end user valued features and benefits.
- Reduced risk of adopting new technology when a wide range of suppliers support the same technology with common specifications.
- Improved product quality and reliability as manufacturers focus on product quality improvement and volume production.

Benefits to Server System Manufacturers

A common specification for server power supplies and electronics bays allows the server manufacturer to design or procure a server chassis with greater assurance of longevity across multiple generations of microprocessor technology. Chassis longevity will allow a system manufacturer to shift a portion of today's R&D budget toward greater value-add efforts rather than repackaging existing technology.

TABLE 1. PLANNED SSI SPECIFICATIONS

Server System Infrastructure Specifications		
Server Power Supplies	*Server Design Guides*	*Server Electronics Bays*
DPS 48V Single Output Distributed		**HEB** ~800 to 1200W Capacity
HPS 750W 5-output	High End Systems 4-8 Processors	**MEB** ~400W Capacity
MPS 375W or 450W 5-output	Midrange Systems 2-4 Processors	**EEB** ~200W Capacity
EPS 300W 5-output	Entry Systems 1-2 Processors	
D2D DC to DC Converter		

An industry specification for electronics bays and power supplies increases the availability of vendors offering competitive products. A wider base of products to select from provides greater flexibility in a make vs. buy decision. A system manufacturer can choose to develop or procure a mix of chassis, boards and power supplies that all work together minimizing system integration time and expense.

Common power supply specifications allow the industry to achieve manufacturing efficiencies and quality improvements based on volume production. Common electronics bays allow chassis suppliers to create off the shelf products that will accept off the shelf power supplies and server electronics based on the SSI specifications.

Benefits to Industry Suppliers

The SSI specifications offer power supply, chassis and server board suppliers a wider range of potential customers. Suppliers can develop products for broad industry application. This reduces the risk associated with development of custom products for specific customers. Time and cost expensed in repackaging technology to fit a customers unique format can be better spent on feature enhancement, quality improvement and manufacturing efficiencies. Suppliers can build their competitive market advantage based on volume production efficiencies and quality improvement of common server elements as compared to competition based on winning new custom designs. SSI also provide for differentiation possibilities without having to compromise the base specification.

II. INITIATIVE HISTORY, ORGANIZATION AND PARTICIPATION

In late 1997, Intel approached the server OEM industry with the concept of SSI and garnered the support of several leading OEMs to establish SSI as an industry program. A governing body called the Promoters' Council was formed

to review draft specifications, to ratify or adopt specifications for public release, and to approve changes to adopted specifications. Today, the Promoters' Council consists of Intel Corporation, Compaq Computer Corporation, Data General Corporation, Dell Computer Corporation, Hewlett-Packard Corporation, NEC, Micron Electronics Incorporated, and Silicon Graphics Incorporated.

Additionally, both server OEMs and suppliers of server hardware elements were invited to join the initiative as Contributors, agreeing to provide non-restrictive or non-licensed feedback to the specifications under development. Today, over 90 companies worldwide participate in SSI as Contributors.

Finally, server OEMs and IHVs have the opportunity to join the SSI initiative as Adopters, under which they are able to use adopted open-industry SSI specifications in product development and design, under royalty-free, cross-license with the Promoters' Council and other Adopters.

Following initial organizational and administrative endeavors, the Promoters' Council focused its energies on development and adoption of the server power subsystem specifications, and released those specifications for public review and Adopter use in early 1999.

Adopted specifications are available for public review at www.ssiforum.org, and may be used for product development and design following the execution of an SSI Adopter's Agreement, also available on the website. As added benefit, Adopters receive advance notification of changes to specifications, notification of compliance tools and events, and identification and promotion as an Adopter company on the SSI website.

Additionally, draft specifications under review by the Promoter's Council are available at the SSI website in a secure area accessible to any company following the execution of an SSI Contributor's agreement.

Feedback by the industry to both adopted and draft specifications is highly encouraged by the SSI Promoters' Council. In order to ensure continued open license of ideas incorporated into the SSI specifications, feedback can only be accepted from SSI Contributors either directly to a member of the Promoters' Council or via the SSI website.

III. Overview of Power Supply Specifications

Five SSI power supply specifications have been adopted for public release and use by Adopters, as outlined in Table 1.

All of the specifications include form factor and other mechanical requirements, connector requirements, current output requirements, chassis and electronics bay interaction requirements, RASUM requirements, as well as safety and environmental requirements. Other requirements exist which are particular to a specific specification. Additionally, feature and functional options and recommendations are included.

EPS – 300W Power Supply for Entry Server Systems

This power supply is intended to be a high-volume, less feature-rich component for entry level servers containing 1-2 Intel® iA32™ processors. Its form factor and chassis interface are identical to the PS2/ATX form factor and its two power and one signal connectors are a superset of the existing ATX connectors. The connector lead lengths are identical to those specified by ATX. The connector scheme will continue to allow non-SSI ATX compliant power supplies to be used with server boards under limited power configurations.

The power supply specifies five outputs consisting of 3.3V, 5V, 12V, -12V, and 5V standby, with total output of 300W.

Airflow characteristic requirements are specified by use of a PQ curve. This power supply is intended to provide air as a cooling source for the server system.

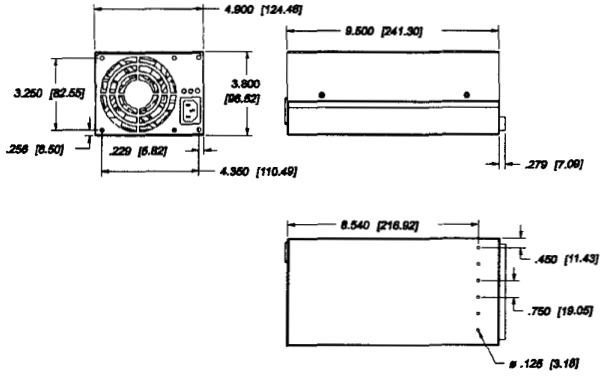
RASUM features specified include 50,000 hours MTBF; fan tachometer predictive failure; I^2C server management medium; ATX specified fan control; and IPMI compliant FRU data. Currently, this specification does not require any redundancy features or capabilities; however, consideration for a redundant capable entry level power supply SSI specification is currently underway.

MPS – 375W or 450W Power Supply for Midrange Server Systems

This power supply is also intended to be a high-volume product, but much more feature rich than the EPS supply. Internal powering sharing is required and the supply must be capable of operating in a redundant $n+1$ configuration. Its output power, specified as either a total of 375W or 400W will support up to four Intel® iA32™ microprocessor systems or dual Intel® iA64™ microprocessor systems. RASUM and chassis/e-bay interaction features are also specified.

The mechanical dimensions of this power supply, shown in Fig. 1, were specified to support a supply that can be oriented in a standard rack bay in a three-across or four-across configuration depending upon the orientation

FIG. 1, MPS MECHANICAL OVERVIEW



of the supply. An optional handle and power supply faceplate are also specified. The handle provides docking and undocking force to mate the supply connector in the system, and also supplies chassis and supply retention.

This supply is not intended to be a primary source of server system cooling.

The MPS supply, rather than its connectors, floats in the chassis as its connector is fixed in the power supply. The connector is specified as a power-blade type with separate signal pins.

RASUM features for this supply are essentially the same as the EPS supply with the exception of additional predictive failure, redundancy, hot swap and LED features.

HPS – 750W Power Supply for High-end Server Systems

Similar to the MPS supply, the HPS supply provides for high availability and high capacity to meet the significant power needs of high-end server systems. It is specified for current output on five rails totaling 750W, which in redundant *n+1* configurations will support systems containing up to eight Intel® iA32™ processors or four Intel® iA64™ processors. It is required to support redundant and hot-swap features.

With the exception of the increased current output, the electrical requirements for this supply are essentially equivalent to those of the MPS supply in order to maintain design consistency. Mechanical specifications are different in form factor only to support the increased volume necessary for greater current output. This supply is specified at 5" wide by 5.3" high by 11 ½" long, which allows for a three-across configuration in a standard rack bay width. The connector for this supply is identical to that specified for the MPS supply. RASUM features are also identical.

DPS – 48V Single-output Distributed Power Supply

This supply represents what is believed to be the introduction into the future of server power. While single-output distributed power schemes have been utilized in very high-end specialized systems, to date the use of distributed power has been widespread in the server industry.

The future's power supply challenges in the server market include an increasing number of voltage rails, including lower voltage rails; higher di/dt; hot swap silver box complexity and increases in distribution lengths and quantity of loads. Distributed power appears to be most able to handle these and other challenges. See Fig. 2 for an overview of a typical distributed power system in a server configuration.

The use of distributed power provides substantial benefits to power delivery in server systems. These benefits include a simplified AC power system; simplified power distribution to the point of load; power capacity which is sized to a specific system configuration; a power system which can be "remote" to other server components; and components which are much more readily reusable.

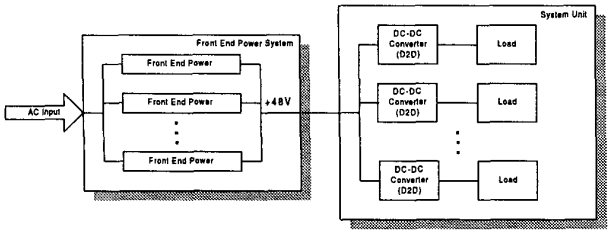
Re-use benefits include a power supply which can be used in any number of different integrated and modular system configurations to power many different server elements; DC-to-DC converters which can be used at multiple points of load; and a power bay concept which can be used to supply any number of modular rack configurations.

Distributed power provides a "pay-as-you-go" power scheme that is very extensible and can be upgraded as system needs demand. Using distributed power in greater numbers of server configurations should also result in higher volume, higher quality, more efficient research and development, and the ability to continue to broaden its application.

The DPS power supply is specified at 800W of current output divided between 750W of 48V power and 50W of 48V standby. It will have configurable capacity to support up to eight Intel® iA32™ processors or up to four Intel® iA64™ processors and multiple bays in a server rack system. RASUM and chassis/e-bay interaction requirements similar to those in the MPS an DPS specifications are also included.

Internal power sharing and hot-swap features are required. A smaller combination power blade and signal pin connector is specified.

FIG. 2: CONCEPTUAL VIEW OF A DISTRIBUTED POWER SYSTEM



D2D – DC-to-DC Converters

Equally integral to a distributed power delivery scheme is a dc-to-dc converter building block. Providing local DC-to-DC conversion provides a solution which is optimized for di/dt requirements; manages the transient load locally; simplifies the power distribution network; and reduces the overall cost of power distribution. The offline switcher solution is simplified and is more loosely coupled to product generation.

The SSI specification for DC-to-DC converters requires models for four voltage outputs: 2.5, 3.3, 5 and 12 with a common 48V input and 35-125W output (. The converters must have current sharing capability built-in. Efficiency must be greater than or equal to 85% in order to meet system cooling requirements. Additionally, the converters must be transformer coupled.

While future generations of SSI specified converters will undoubtedly require smaller form factor footprints and concurrent connector changes, the current specification calls for dimensions not to exceed 1.9" high by 3.8" long by 1.1" wide . As currently specified, the D2D utilizes a 60-pin pin and socket type connector.

Safety and Environmental Requirements

The safety and environmental requirements for SSI power supplies are identical and consistent among specifications.

Safety requirements include an AC Input Range of 100-240VAC 50/60Hz; UL, CSA and CE certification; and a CB report.

Environmental requirements include power factor correction (PFC) as well as compliance with FCC/ICES-003 (emissions), CISPR 22/EN55022 (emissions), EN50082-1 (immunity), EN61000-3-2 (harmonics), and CE requirements.

IV. THE FUTURE OF SSI

The SSI Promoters' Council will continue to examine the appropriateness, accuracy and applicability of adopted specifications, and will entertain suggestions for revision from the entire industry. It is anticipated that subsequent revisions of all adopted specifications will be considered and adopted as change to the specifications becomes necessary. All changes to adopted specification will require the same examination and scrutiny as initially given to adoption, and will require a majority vote of the Promoters' Council. The SSI Promoters' Council strongly urges and requests the ongoing review and feedback of APEC and similar entities, as well as individual companies, in order to ensure that the SSI specifications have broad and easy use and application with considerable longevity.

It is also anticipated that additional power and other specifications will be considered for future adoption. Among those under consideration of interest to APEC are an entry-level redundant power supply and a common power distribution backplane.

ACKNOWLEDGMENT

Intel would like to acknowledge the significant cooperation and support of the Promoters' Council by Celestica, Inc. and Delta Electronics, Inc. in the development and review of the power supply specifications; as well as the many other SSI Contributor companies who provided thoughtful and productive feedback to the developing SSI specifications.